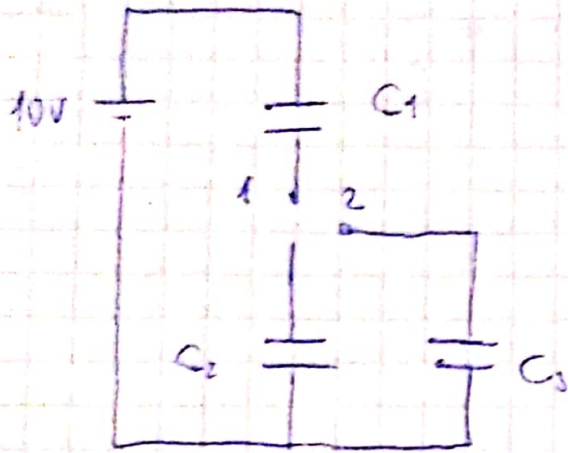


Ejercicio 5 - GN



$$C_1 = 12 \mu F$$

$C_3 = 2 \mu F$ descargado

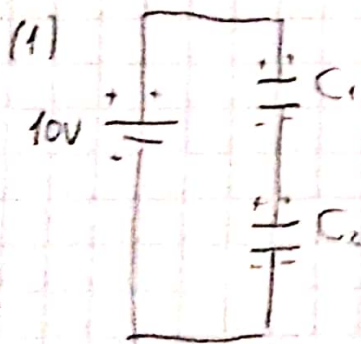
$$C_2 = 6 \mu F$$

Se carga en (1), descarga en (2)

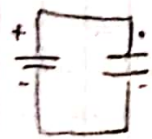
a) $Q_{F3}?$

b) Energía de $C_1 - C_2$ antes de conectar

la llave y por C_2, C_3 después de hacer



$$C = \frac{Q}{V}$$



$$\rightarrow \frac{1}{C_{1,2}} = \frac{12 \cdot 6}{18} = 4 \mu F$$

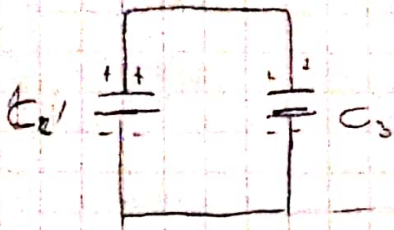
$$\Rightarrow C_s = \frac{Q_T}{V}$$

$$Q_T = C_{1,2} \cdot V = 4 \mu F \cdot 10V = 40 \mu C$$

$$Q_T = Q_1 = Q_2$$

↳ por conexión serie

(2)



$$\textcircled{1} \quad \overbrace{q_{2F}}^{40 \mu C} = q_{2F} + q_{3F} \quad (1)$$

$$\textcircled{2} \quad U_{2F} - U_{3F} = 0 \quad (2)$$

$$(2) \quad U_{2F} = U_{3F}$$

$$\frac{q_{2F}}{C_2} = \frac{q_{3F}}{C_3}$$

$$\frac{40 \mu C - q_{3F}}{C_2} = \frac{q_{3F}}{C_3} \rightarrow \begin{cases} q_{3F} = 10 \mu C \\ q_{2F} = 30 \end{cases}$$

$$\Rightarrow U_F = \frac{30}{6} = 5V$$

$$\textcircled{b} \quad E_{1,2} = \frac{C V^2}{2} = \frac{4 \mu F \cdot (10V)^2}{2} = \frac{4 \cdot 10^{-6} \cdot 100V^2}{2} = 200 \mu J$$

$$E_{1,2,3} = ?$$